

Building and Evaluating Models with Your Labeled Data

Dataset Inventory and Annotation: First, **profile your labeled data** to understand its structure and quality. Check the number of samples per class, class balance, missing or duplicate entries, and data formats. Maintain **comprehensive documentation** of your data sources, transformations, and labeling process ¹. For example, record how many images belong to each category and note any anomalies. Use **data profiling techniques** (histograms, summary stats) to spot inconsistencies or outliers ². Inspect your annotation schema: ensure labels are well-defined and consistently applied. If working with images, document whether labels are simple class tags or include bounding boxes/segmentation. Adopt a standard format (like COCO) and clearly specify label definitions and edge-case rules ³. This inventory step lays the groundwork for reliable modeling.

Baseline Models (Image Classification & Forensics): Next, select simple **baseline models** to set performance benchmarks. For general image classification, common choices include pre-trained CNN backbones (ResNet, VGG, EfficientNet, etc.) fine-tuned on your data. Emerging **Vision Transformers (ViTs)** are also popular for high-dimensional image tasks. In a forensic context (e.g. detecting manipulated or AI-generated images), specialized networks are often used. For instance, the Xception architecture (with depthwise separable convolutions) is a strong baseline in deepfake detection ⁴. In one study, Xception, ResNet-50, VGG16 and MobileNetV2 were evaluated: Xception achieved the highest accuracy (89.2%), while VGG16 was best for precision and ResNet for speed ⁴. These models serve as starting points. You can also explore lightweight or ensemble approaches as needed. Transfer learning from ImageNet weights is usually effective: e.g. replacing the final layers of a pre-trained CNN to match your classes.

Experiment Design (Cross-Validation & Class Imbalance): When designing experiments, ensure robust evaluation. Use **k-fold cross-validation** (e.g. k=5 or 10) to assess generalization. **Stratified** splits are crucial for classification: they preserve the class proportions in each fold, which prevents evaluation bias on imbalanced data ⁵. Imbalanced classes often occur in practice; for example, some product categories or rare forgeries might be underrepresented. Apply class balancing techniques: you can oversample minority classes (or undersample majority) in training, or use class-weighting in the loss function to compensate for skew. For example, oversampling or using the [sklearn class_weight](#) option ensures the model sees enough examples of each class. Cross-validation folds should **reflect realistic class ratios** so performance estimates match real-world use.

Evaluation Metrics: Choose metrics that match your goals and data balance. For classification, report **precision, recall, and F1-score** for each class (especially important if classes are imbalanced). Also compute **ROC AUC** (area under the ROC curve) when appropriate: it measures the trade-off between true and false positives across thresholds, which is valuable for binary or multi-class problems ⁴. Matthews Correlation Coefficient (MCC) is another robust metric for imbalanced data. In forensic tasks (real vs. fake images), precision and recall are critical: a detector must minimize false positives (mislabeling real images as fake) as well as false negatives.

Beyond standard metrics, assess **calibration** of predicted probabilities. A well-calibrated model's confidence scores match the true likelihood of correctness. For trustworthiness, "a model must be calibrated so that its confidence in each decision closely reflects its true outcome" ⁶. You can measure

calibration error (e.g. Expected Calibration Error, ECE) or produce reliability diagrams. For instance, one study found Xception and VGG16 had much lower ECE (better calibration) than ResNet-50 on a deepfake task ⁷. Well-calibrated models are especially important when probabilities inform downstream decisions (e.g. flagging suspicious listings). Always review the confusion matrix to see which types of errors occur.

Deployment, Drift, and Adversarial Testing: Finally, plan for **real-world deployment and robustness**. In production, the data can change over time (data drift or concept drift). *Data drift* refers to changes in the input distribution after deployment; as one guide notes, “the real-world data starts to differ from the training data, breaking the model’s assumptions and making predictions less accurate” ⁸. To monitor this, implement drift detectors: compare feature distributions over time (e.g. use statistical tests like Kolmogorov-Smirnov or Population Stability Index) and track key input statistics. Also regularly check model performance metrics on new labeled data or proxy signals. If drift is detected, schedule retraining with recent data.

Consider **adversarial robustness** as well. Forensic detection models (e.g. forgeries) can be vulnerable to small perturbations. Test your model by simulating adversarial examples (e.g. FGSM or PGD attacks) to see how accuracy degrades. Research has shown that even high-accuracy detectors “exhibit reduced performance under adversarial conditions, underscoring the need for enhanced resilience” ⁴. Incorporate adversarial testing in validation, and consider defenses (like adversarial training or data augmentation) if feasible.

By systematically inventorying your dataset, choosing clear annotation standards, selecting and benchmarking baseline models, carefully designing your evaluation (with stratification and proper metrics), and planning for drift and attacks, you build a thorough framework. This ensures your model is reliable, interpretable, and maintains performance even as conditions change.

Sources: We draw on best practices and recent research. For instance, data-quality guidelines emphasize thorough documentation and profiling ¹ ². Annotation resources recommend clear schemas and standard formats ³. Deep learning surveys and experiments illustrate how CNN backbones (ResNet, VGG, Xception) perform on image and forensic tasks ⁴. Guidance on evaluation covers stratified cross-validation for imbalanced data ⁵, and highlights precision/recall/AUC ⁴ and calibration concepts ⁶. Finally, ML-in-production literature discusses data drift detection ⁸ and the importance of adversarial testing ⁴. These combine to inform a comprehensive modeling plan.

¹ ² Best Practices for Data Quality in Machine Learning

<https://www.anomalo.com/blog/data-quality-in-machine-learning-best-practices-and-techniques/>

³ Why Dataset Annotation is Key to Machine Learning | Voxel51

<https://voxel51.com/blog/why-quality-dataset-annotation-is-key-to-machine-learning>

⁴ ⁷ (PDF) Comprehensive Evaluation of Deepfake Detection Models: Accuracy, Generalization, and Resilience to Adversarial Attacks

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/388436656_Comprehensive_Evaluation_of_Deepfake_Detection_Models_Accuracy_Generalization_and_Resilience_to_Adversarial_Attacks)

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⁵ How to Fix k-Fold Cross-Validation for Imbalanced Classification - MachineLearningMastery.com

<https://machinelearningmastery.com/cross-validation-for-imbalanced-classification/>

⁶ Understanding Model Calibration - A gentle introduction and visual exploration of calibration and the expected calibration error (ECE)

<https://arxiv.org/html/2501.19047v2>

8 Detecting & Handling Data Drift in Production - MachineLearningMastery.com
<https://machinelearningmastery.com/detecting-handling-data-drift-in-production/>